DIOLL, A FLUORESCENT PROBE FOR IN-DEPTH CHARACTERIZATION OF CELL MEMBRANE FLUIDITY

Quentin VALLEE[1,2,3*], Sébastien IBANEZ[2,4*], David GUEYRARD[4], Enzo AMOROSO[2], Mouhedine CHENIOUR[4], Larissa SOCRIER[4], Jihen ZOUAOUI[4,5], Samia SELMI-RUBY[2], Thomas GUILLOT[2], Anna BEROT[4,6], Sifre VAN TEEFFELEN[7], Ahmed LANDOULSI[5], Agnès GIRARD-EGROT[4], Christophe VANBELLE[2], Meddy EL ALAOUI[6], Carlos MARQUES[7], Peter GOEKJIAN[4], Ofélia MANITI[4], Olivier MARCILLAT[2], Thierry GRANJON[4], Arnaud M. VIGNERON[2,3,8*]

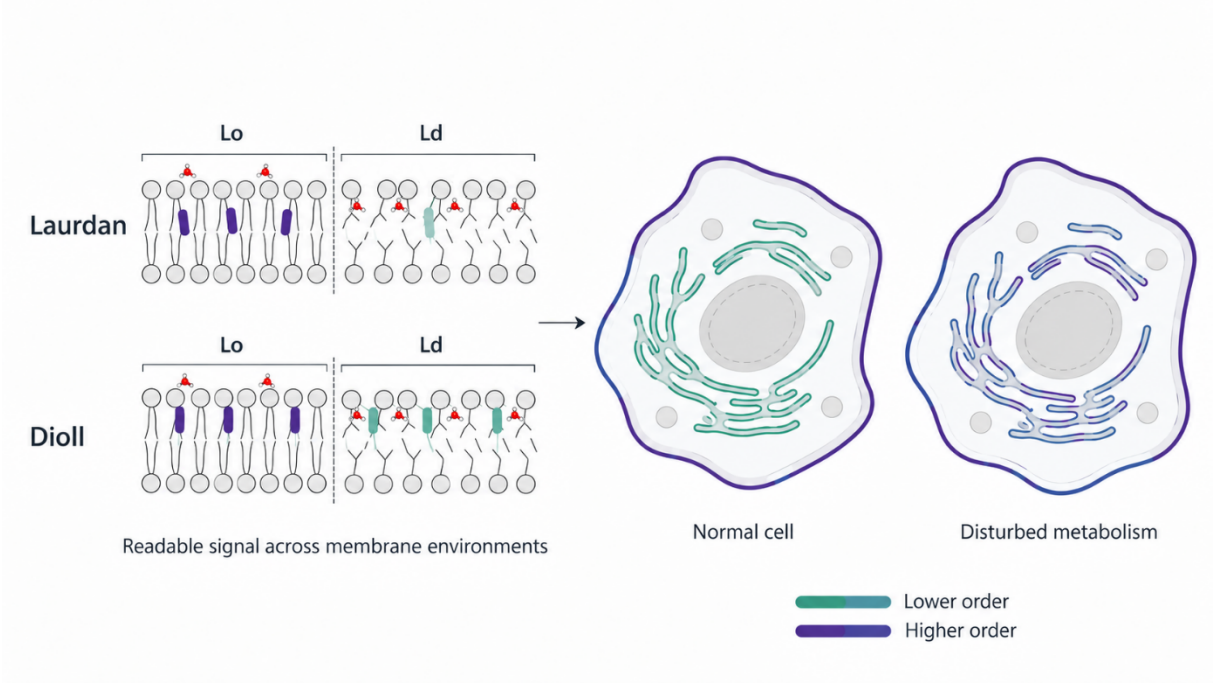
[1] University Grenoble-Alpes, Ecole doctorale Chimie-Science du vivant, Grenoble, France; [2] University of Lyon, Centre de Recherche en Cancérologie de Lyon, UMR INSERM 1052 CNRS 5286 - Centre Léon Bérard - 28 Rue Laennec, Lyon, France; [3] Sirius NeoSight - Centre Léon Bérard - 28 Rue Laennec, Lyon, France; [4] University of Lyon, Laboratoire Chimie Organique 2, ICBMS UMR 5246 CNRS-UCBL-CPE, Université Lyon 1, Villeurbanne, France; [5] Laboratoire de Biochimie et de Biologie Moléculaire, Unité Risques Liés au stress Environnemental, lutte et prévention UR17ES20, Faculté des sciences de Bizerte, Jarzouna, Bizerte, Tunisie; [6] Covalab S.A.S., Europarc du Chêne, 1 rue Jacques Monod, 69500 Bron, France; [7] University of Lyon, ENS-Lyon, CNRS UMR 5182, Chem. Lab, Lyon 69342, France, [1] University Grenoble-Alpes, Institut pour L'Avancée des Biosciences, UMR INSERM 1209 CNRS 5309, Site Santé, Allée des Alpes, La Tronche, France;

Keywords: Membrane lipid order; Membrane fluidity; Solvatochromic fluorescent probe; Laurdan; Live-cell imaging; Membrane heterogeneity; Metabolic regulation.

ABSTRACT

Membrane lipid order contributes to the organization of cellular compartments and the regulation of membrane-associated processes, yet its spatial heterogeneity and dynamic changes remain difficult to assess in living systems.[1] Environment-sensitive probes such as Laurdan and C-Laurdan allow lipid order to be estimated from fluorescence emission, but differences in probe distribution and signal intensity across membrane environments can complicate their interpretation.[2-4] Here, we report the synthesis and characterization of Dioll, a new Laurdan-derived probe designed to improve lipid-order measurements in living cells and tissues. The introduction of a short dihydroxypropyl group increases fluorescence intensity while maintaining sensitivity to changes in membrane hydration and lipid packing. Compared with Laurdan and C-Laurdan, Dioll provides a broader generalized-polarization (GP) response and remains bright in disordered membranes. At suitable staining concentrations, it samples both liquid-ordered and liquid-disordered environments more evenly, while showing little accumulation in lipid droplets. These properties allow us to distinguish lipid-order profiles across plasma and intracellular membranes and to monitor changes induced by lipid composition and cell density. Dioll also enables two-photon imaging of epithelial membranes

in intact mammary tissue, including regions poorly visualized with Laurdan. Together, our results establish Dioll as a sensitive and practical tool for investigating the organization and regulation of membrane lipid order in living cells and tissues.



Graphical abstract. *Unlike Laurdan, Dioll provides strong fluorescence in both liquid-ordered (Lo) and liquid-disordered (Ld) membranes, enabling comprehensive mapping of membrane lipid order in living cells and its modulation by metabolic perturbations.*

REFERENCES

(1) Sezgin, E.; Levental, I.; Mayor, S.; Eggeling, C. The mystery of membrane organization: composition, regulation and roles of lipid rafts. Nat. Rev. Mol. Cell Biol. 2017, 18, 361–374. https://doi.org/10.1038/nrm.2017.16

(2) Parasassi, T.; De Stasio, G.; Ravagnan, G.; Rusch, R. M.; Gratton, E. Quantitation of lipid phases in phospholipid vesicles by the generalized polarization of Laurdan fluorescence. Biophys. J. 1991, 60, 179–189. https://doi.org/10.1016/S0006-3495(91)82041-0

(3) Kim, H. M.; Choo, H.-J.; Jung, S.-Y.; et al. A two-photon fluorescent probe for lipid raft imaging: C-Laurdan. ChemBioChem 2007, 8, 553–559. https://doi.org/10.1002/cbic.200700003

(4) Gunther, G.; Malacrida, L.; Jameson, D. M.; Gratton, E.; Sánchez, S. A. LAURDAN since Weber: The quest for visualizing membrane heterogeneity. Acc. Chem. Res. 2021, 54, 976–987. https://doi.org/10.1021/acs.accounts.0c00687